

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2147

Hancock 345 kV
July 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2147. The Interconnection Customer proposes a 726.2 MW natural gas combined cycle facility interconnecting at the Hancock 345 kV substation in the Virginia area. The request was received on July 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 731 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,262,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Starkey - Hancock 138.0 kV
Contingency: Reusens - Huntington Court 345.0 kV L
Loading: 151.7% (Rating: 310 MVA)
DFAX: 25.1%
Pre-project loading: 92.9%
2. Facility: Hancock 345/138kV Auto
Contingency: Huntington Court - Hancock 345.0 kV
Loading: 141.3% (Rating: 514 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
3. Facility: Meadowview - Poages Mill 138.0 kV
Contingency: Reusens - Huntington Curt 345.0 kV
Loading: 136.3% (Rating: 267 MVA)
DFAX: 18.8%
Pre-project loading: 85.2%
4. Facility: Huntington Court 345 kV/138 kV XFMR
Contingency: Huntington Court - Hancock 345.0 kV line
Loading: 133.5% (Rating: 500 MVA)
DFAX: 30.9%
Pre-project loading: 88.6%
5. Facility: Jack Ramey Wind - Starkey 138.0 kV
Contingency: Reusens - Huntington Court 345.0 kV
Loading: 127.0% (Rating: 342 MVA)
DFAX: 25.1%
Pre-project loading: 73.8%

6. Facility: Jack Ramey Wind Substation - Clearbrook 138.0 kV
Contingency: Reusens - Huntington Court 345.0 kV
Loading: 111.8% (Rating: 366 MVA)
DFAX: 25.1%
Pre-project loading: 62.1%
7. Facility: Rally Hill - Meadowview 138.0 kV
Contingency: Huntington Court - Reusens 345.0 kV
Loading: 109.1% (Rating: 300 MVA)
DFAX: 18.8%
Pre-project loading: 63.6%
8. Facility: Huntington Court - Steamboat 138.0 kV
Contingency: Hancock - Huntington Court 345.0 kV
Loading: 106.1% (Rating: 357 MVA)
DFAX: 29.7%
Pre-project loading: 45.5%
9. Facility: Florida - N Huntington 138.0 kV
Contingency: Craven County Wood Energy Lp - Lone Tree 138.0 kV
Loading: 106.1% (Rating: 577 MVA)
DFAX: 10.4%
Pre-project loading: 93.1%
10. Facility: Berlin Gorham - Florida 138.0 kV
Contingency: Craven County Wood Energy Lp - Lone Tree 138.0 kV
Loading: 106.0% (Rating: 584 MVA)
DFAX: 10.4%
Pre-project loading: 93.1%
11. Facility: Skiffes Creek - Berlin Gorham 138.0 kV
Contingency: Craven County Wood Energy Lp - Lone Tree 138.0 kV
Loading: 105.7% (Rating: 597 MVA)
DFAX: 10.4%
Pre-project loading: 93.1%
12. Facility: Michigan Power Lp - Skiffes Crk 138.0 kV
Contingency: Craven County Wood Energy Lp - Lone Tree 138.0 kV
Loading: 105.4% (Rating: 613 MVA)
DFAX: 10.4%
Pre-project loading: 93.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Starkey - Hancock 138.0 kV	Rebuild line with higher-capacity conductor	\$2,798,000	32% (\$902,000)
Hancock 345/138kV Auto	Replace transformer with higher-rated unit	\$15,145,000	100% (\$15,145,000)
Meadowview - Poages Mill 138.0 kV	Rebuild line with higher-capacity conductor	\$4,336,000	30% (\$1,304,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Huntington Court 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$19,990,000	26% (\$5,225,000)
Jack Ramey Wind - Starkey 138.0 kV	Rebuild line with higher-capacity conductor	\$3,510,000	32% (\$1,132,000)
Jack Ramey Wind - Clearbrook 138.0 kV	Rebuild line with higher-capacity conductor	\$1,255,000	32% (\$405,000)
Rally Hill - Meadowview 138.0 kV	Reconductor line	\$1,515,000	30% (\$455,000)
Huntington Court - Steamboat 138.0 kV	Reconductor line	\$1,854,000	37% (\$684,000)
Flouda - N Huntington 138.0 kV	Reconductor line	\$2,709,000	10% (\$283,000)
Berlin Gorham - Flouda 138.0 kV line	Reconductor line	\$1,309,000	10% (\$137,000)
Skiffes Creek - Berlin Gorham 138.0 kV	Reconductor line	\$4,030,000	10% (\$420,000)
Michigan Power Lp - Skiffes Crk 138.0 kV	Reconductor line	\$1,627,000	10% (\$170,000)
	Total allocated to project		\$26,262,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.